

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE CODE: CSA51**

**COURSE NAME: Cryptography and Network Security**

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### **COURSE OUTCOME COVERED**

**CO5:** *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

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**QUESTIONS: 5**

**TOTAL MARKS: 25**

### **Annexure A**

#### **PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)**

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#### ***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## Annexure A

### PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.
2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.
3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.
4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.
5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

#### RUBRICS FOR CALCULATION:

| Q. No                        | Understanding of Concepts (25) | Experimental Design (30) | Use of Tools (20) | Analysis of Results (15) | Documentation (10) | Total Score (out of 100) |
|------------------------------|--------------------------------|--------------------------|-------------------|--------------------------|--------------------|--------------------------|
| 1                            | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
| 2                            | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
| 3                            | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
| 4                            | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
| 5                            | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
| <b>OVERALL RUBRICS VALUE</b> |                                |                          |                   |                          |                    |                          |
|                              | / 4                            | / 4                      | / 4               | / 4                      | / 4                |                          |
|                              | / 25                           | / 30                     | / 20              | / 15                     | / 10               | / 100                    |

## **CO5 AT2 CRYPTO:**

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### **1. Simplified TLS Handshake**

#### **Explanation:**

The TLS handshake establishes a secure connection between a client and server. The client sends a request, the server provides its digital certificate for authentication, both sides generate a session key, and then encrypted data is exchanged.

PSEUDOCODE: SIMPLIFIED TLS HANDSHAKE

BEGIN

Client → Server: Send connection request

Server → Client: Send digital certificate

Client:

    Verify server certificate

    IF certificate is valid THEN

        Continue handshake

    ELSE

        Reject connection

    STOP

    END IF

Client and Server:

    Generate/establish a shared session key

Client → Server:

    Send encrypted application data

Server:

Decrypt received data

Process the data

Server → Client:

Send encrypted response

Client:

Decrypt and receive response

Display "Secure communication established"

END

**Flow:**

Client Request → Certificate Verification → Session Key Generation → Encryption →  
Secure Data Exchange

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## 2. Firewall Rule Engine

**Explanation:**

A firewall checks incoming packets against predefined rules. The rules can contain an IP address, port number, and protocol. If a packet matches an allowed rule, it is accepted; otherwise, it can be blocked.

PSEUDOCODE: FIREWALL RULE ENGINE

BEGIN

Load firewall rules

WHILE packet is received DO

    Read source IP

    Read destination IP

    Read port number

    Read protocol

    matched  $\leftarrow$  FALSE

    FOR each rule in firewall rules DO

        IF IP matches rule AND

        port matches rule AND

        protocol matches rule THEN

            matched  $\leftarrow$  TRUE

        IF rule action = ALLOW THEN

            Allow packet

            Log "Packet Allowed"

        ELSE

            Block packet

            Log "Packet Blocked"

        END IF

    BREAK

END IF

END FOR

IF matched = FALSE THEN

Block packet

Log "No matching rule - Packet Blocked"

END IF

END WHILE

END

**Flow:**

Incoming Packet → Check IP → Check Port → Check Protocol → Match Rule → Allow/Block

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### 3. Malware Behavior Classification

**Explanation:**

Malware can be classified by observing how it spreads and behaves. A **virus** commonly modifies or attaches to files, a **worm** spreads automatically through networks, and a **Trojan** usually depends on the user running or installing something that appears legitimate.

PSEUDOCODE: MALWARE CLASSIFICATION

BEGIN

Read malware behavior

Check propagation method

Check file modification pattern

Check user interaction

IF malware modifies or attaches to files AND  
requires a host file THEN

Classification ← "Virus"

ELSE IF malware spreads automatically through  
networks or connected systems THEN

Classification ← "Worm"

ELSE IF malware requires user interaction such as  
opening or installing a seemingly legitimate program THEN

Classification ← "Trojan"

ELSE

Classification ← "Unknown Malware"

END IF

Display Classification

END

**Flow:**

Analyze Behavior → Check Propagation → Check File Modification → Check User  
Interaction → Classify Malware

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#### 4. Intrusion Detection System (IDS)

**Explanation:**

An IDS monitors network activity and looks for suspicious behavior. **Signature-based detection** compares activity with known attack patterns, while **anomaly-based detection** compares activity with normal network behavior.

PSEUDOCODE: IDS

BEGIN

Load known attack signatures

Establish normal network behavior

WHILE network traffic is received DO

Capture network packet

signature\_match ← Check packet against known signatures

anomaly\_score ← Compare packet with normal behavior

IF signature\_match = TRUE THEN

Alert ← "Known Attack Detected"

Log intrusion

ELSE IF anomaly\_score > threshold THEN

Alert ← "Abnormal Activity Detected"

Log intrusion



ELSE

Alert ← "Normal Traffic"

END IF

Display Alert

END WHILE

END

**Flow:**

Network Traffic → Signature Check + Anomaly Check → Suspicious? → Generate Alert → Log Event

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## 5. Intrusion Prevention System (IPS)

**Explanation:**

An IPS works in real time and not only detects attacks but also takes action. When malicious traffic is identified, the IPS can block the request and record the intrusion in a log.

PSEUDOCODE: IPS

BEGIN

Load security rules

Load known attack signatures

Set anomaly threshold

WHILE network traffic is received DO

Capture request

Check request against attack signatures

Calculate anomaly score

IF known malicious signature is detected OR  
anomaly score > threshold THEN

Block request

Log:

Source IP

Destination IP

Protocol

Time

Attack type

Generate security alert

ELSE

Allow request

Log "Normal Traffic"

END IF

END WHILE

END

**Flow:**

Real-Time Traffic → Analyze Request → Detect Malicious Activity → **Block + Log + Alert** / Allow

**Quick Difference**